

Self-Supervised and Generative Learning Approaches for EEG-Based Sleep Stage Classification

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Abstract—Sleep staging from EEG signals is an essential step in diagnosing sleep disorders, yet remains challenging due to class imbalance, especially for N1 sleep. In this work, we evaluate four deep learning architectures on the Sleep-EDF Sleep Cassette dataset: a supervised 1D CNN baseline, a CNN with Variational Autoencoder (VAE), a self-supervised SimCLR-based approach, and a hybrid SSL+VAE model. Results show that the baseline 1D CNN achieves the highest accuracy of 78.07%, while the VAE-augmented model substantially improves the recall of the underrepresented N1 stage (from 11.96% to 27.35%). Pure self-supervised learning underperforms due to class collapse, achieving only 71.19% accuracy and nearly failing on N1. The hybrid SSL+VAE model partially mitigates this, recovering performance to supervised levels. These findings demonstrate that generative latent space shaping contributes more to minority-stage learning than contrastive learning alone.

Index Terms—Sleep Staging, EEG, Self-Supervised Learning, Variational Autoencoder, Deep Learning

I. INTRODUCTION

Automatic sleep staging enables scalable sleep analysis and early diagnosis of neurological and physiological disorders. The standard approach relies on manual annotation of polysomnography (PSG) recordings into sleep stages, which is time-consuming and costly. Machine learning methods, particularly deep learning, have shown strong performance in automated sleep staging; however, they struggle with class imbalance, especially in the N1 stage, which typically represents less than 10% of total sleep.

Recent advances in representation learning, including Variational Autoencoders (VAEs) and contrastive Self-Supervised Learning (SSL), demonstrate potential for learning richer feature embeddings without requiring large labeled datasets. In this work, we compare supervised, generative, and self-supervised learning approaches for sleep staging using EEG from the Sleep Cassette portion of the Sleep-EDF database.

II. METHODS

A. Dataset

We use the Sleep-EDF Expanded Sleep Cassette dataset, consisting of 20 subjects and overnight recordings sampled at 100 Hz. Sleep stages are labeled as: W, N1, N2, N3, and REM.

B. Models Evaluated

We trained and compared four models:

- **1D CNN Baseline:** A fully supervised convolutional model for raw EEG classification.
- **1D CNN + VAE:** The VAE learns a latent representation jointly trained with the classifier.
- **SimCLR SSL:** A contrastive encoder trained without labels, followed by a linear classifier.
- **SSL + VAE:** Combines latent reconstruction with contrastive learning.

The key objective was to evaluate whether generative and self-supervised learning improve stage-wise performance, particularly for N1.

III. RESULTS

A. Overall Performance

The 1D CNN baseline achieves the highest overall accuracy of 78.07%. The SimCLR model shows the lowest accuracy at 71.19%, confirming that contrastive learning alone does not guarantee strong downstream performance on imbalanced tasks.

B. Per-Class Analysis

Table I shows the f1-scores for the 5 stages:

TABLE I
F1-SCORE COMPARISON ACROSS MODELS

Stage	CNN	CNN+VAE	SSL	SSL+VAE
W	80.00	80.67	41.64	71.23
N1	19.04	27.58	1.47	16.22
N2	83.34	82.48	70.73	75.11
N3	81.95	79.67	69.82	76.45
REM	73.65	70.65	75.42	71.88

C. Key Observations

- The CNN+VAE improves N1 recall from 11.96% to 27.35%.
- SSL almost collapses on N1 ($f1 = 1.47$).
- SSL reduces recall for W from 73.68% to 26.62%.
- Hybrid SSL+VAE recovers performance but remains weaker than full supervision.

IV. DISCUSSION

The results show that while SSL improves feature generalization in many domains, it is insufficient for sleep staging under high class imbalance. The VAE-based generative model meaningfully enhances minority stage detection by regularizing the latent space and improving separability. This suggests that generative regularization is a promising direction for physiological signals where data imbalance is inherent.

V. CONCLUSION

We benchmarked supervised, generative, and self-supervised methods for EEG-based sleep staging. While the supervised 1D CNN remained the top performer, the VAE-enhanced model offered significant gains for minority stage N1. SSL alone suffers from severe class imbalance effects, but partial recovery is observed when combined with VAE. Future work could explore hybrid loss balancing, larger projection heads, and multi-channel input fusion.

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